

LIFD: Anchored Diffusion for 3D-Aware Scene Memory in Robotic Manipulation

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Abstract—Robotic manipulation under partial observability requires spatial information that extends beyond the current view. Geometry-aware RGB features describe visible structure, but previously observed regions may disappear as the robot or scene moves. Maintaining a useful scene representation therefore requires retaining observation history while inferring missing content without losing its connection to visible evidence. We introduce LIFD (Look, Imagine, Focus, and Do), a framework for persistent, 3D-aware scene memory. LIFD learns a scene-token representation from multi-view agreement and completes it from a single RGB view and recurrent memory. A rectified-flow model generates the tokens while Anchor-Guided Cross-Attention conditions completion on current geometric features. Compact slot features connect this representation to a manipulation policy. Multi-view and geometric supervision are used during representation learning; deployment requires one RGB camera, proprioception, and a task instruction. LIFD (Staged) reaches 91.6% average success on LIBERO and 79.8% on MetaWorld, improving LIBERO average success by 3.1 percentage points over Joint training. On four UR5e task families with ten demonstrations per family, it achieves 56.0% mean success, compared with 40.5% for OpenVLA-7B.

I. INTRODUCTION

Robotic manipulation in cluttered, partially observable environments requires a persistent understanding of scene structure. An object can remain relevant to an action after it leaves the current field of view, while manipulation itself changes what the camera observes. The representation used for control must therefore integrate newly visible structure with information accumulated during execution. This requirement concerns both what the robot remembers and how it interprets incomplete visual evidence.

Spatial representations provide useful inductive biases for manipulation, including point clouds and voxels [1]–[3]. Acquiring these inputs introduces depth-sensing or reconstruction requirements. Geometry-aware image encoders offer a route to 3D priors from RGB [4], but the current view still leaves parts of the scene unobserved. Recent memory-based policies already retain observation history and spatial context [5]–[7]. The challenge is to maintain a scene-level representation when its supporting evidence is distributed across incomplete views and time.

Memory and generative completion address different aspects of this problem. Memory carries information from earlier observations, while generative models provide learned priors for inferring unobserved content [8], [9]. A plausible completion can nevertheless conflict with currently visible

structure. For manipulation, inferred scene content must remain tied to visible geometric evidence and information accumulated over time. This motivates a scene representation learned across views, whose completion is conditioned jointly on current observations and temporal context.

We introduce LIFD (Look, Imagine, Focus, and Do), a framework for persistent, 3D-aware scene memory. Its central design is to learn a scene-level representation from multi-view observations, then complete that representation from local RGB evidence and recurrent memory. Self-distillation encourages agreement across views, while anchored rectified flow conditions scene-token generation on current geometry-aware features and temporal context. *Look* and *Imagine* establish this representation; *Focus* and *Do* condense it into slot features and predict actions. The scene tokens form an interface between geometric perception and a lightweight manipulation policy. Multi-view and geometric supervision shape the representation during training, while deployment uses a single camera stream.

Experiments on LIBERO, MetaWorld, RoboTwin 2.0, and four UR5e task families support the control utility of this scene-memory formulation under the evaluated adaptation protocol. Ablations and observation interventions assess how generative completion, visible anchoring, and accumulated history contribute to control, alongside the cost of iterative inference.

II. RELATED WORK

A. Visual Representations for Manipulation

Visual pretraining supplies transferable features for visuomotor learning [10], [11]. Point-cloud and voxel policies expose spatial structure for manipulation [1]–[3]. Geometry-aware image encoders such as VGGT predict camera parameters, depth, point maps, and tracks from image collections [4]. LIFD uses these geometric priors to learn a scene-token representation for RGB deployment, with multi-view supervision available during training.

B. Memory for Robotic Manipulation

Memory-based policies address observation ambiguity through different historical interfaces. TraceVLA overlays tracked point trajectories on current images [12], while MemoryVLA stores perceptual and semantic features for retrieval and fusion [6]. SAM2Act+ extends a 3D manipulation policy with memories of visual features and predicted action heatmaps [5]. BridgeVLA++ combines temporal context

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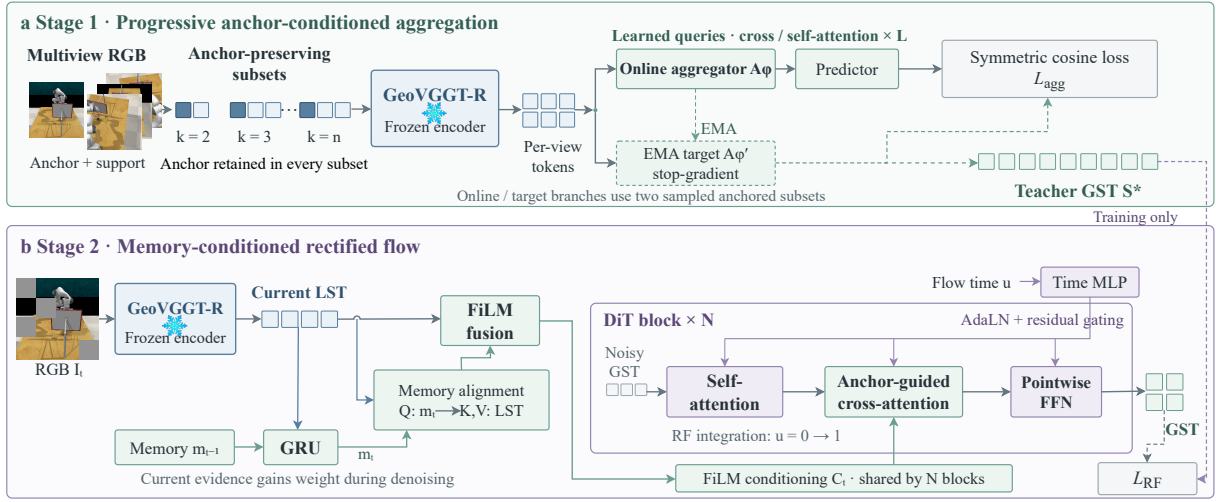


Fig. 1. **Staged representation learning.** (a) Anchor-conditioned aggregation and self-distillation learn multi-view scene tokens. (b) Rectified flow completes them from current evidence and memory, progressing from noise at $u = 0$ to teacher GST at $u = 1$.

with spatial memory constructed from initial point clouds [7]. These approaches already connect spatial perception and observation history for control. LIFD addresses this problem through generative completion: recurrent history and current RGB features jointly condition predictions of a multi-view scene-token target.

C. Generative Representations and Control

World models learn environment representations and dynamics for planning and control [8], [9], [13]. Diffusion-based policies generate actions or trajectories [14], [15], and WorldVLA jointly models action generation and future images [16]. LIFD uses generation to infer a current scene representation for downstream control. Its multi-view targets build on set-latent aggregation [17], [18], while a rectified-flow objective [19] learns conditional completion in token space. The generated scene tokens are conditioned on observation history and visible evidence; they are not action-conditioned future predictions.

III. METHOD

LIFD learns a scene representation from multi-view supervision, then completes that representation from a single RGB stream and its observation history. At environment step t , a geometry-aware encoder E maps the image I_t to Local Sense Tokens (LST), $T_t \in \mathbb{R}^{N \times d}$. Recurrent memory and current LST condition the generation of Global Scene Tokens (GST), $\hat{S}_t \in \mathbb{R}^{M \times d}$, which supply the policy with a scene-level representation. Here N and M are token counts and d is the feature dimension. GST carry learned geometric features without explicit metric coordinates. Proprioception P_t and instruction embedding e_L enter the policy head. Fig. 1 summarizes representation learning; Fig. 2 shows the deployed information flow.

A. Learning a Geometry-Aware Scene Representation

From local geometry to scene-level targets. GeoVGGT-R is obtained by LoRA-adapting a pretrained VGGT back-

bone [4] with camera, depth, point-map, and tracking supervision, and is frozen in subsequent stages. These geometric priors make the RGB features sensitive to spatial structure, while multi-view aggregation supplies information beyond an individual view. The supervision heads and loss definitions are given in Appendix A.

For views $\mathcal{I} = \{I_a, I_2, \dots, I_n\}$, every training subset contains the same anchor image I_a ; the number of support views increases progressively from one to $n - 1$. For an anchor-containing subset $C \subseteq \mathcal{I}$, an aggregator A_ϕ uses learned queries to collect evidence into a fixed-length scene representation:

$$S(C) = A_\phi(\{E(I) : I \in C\}), \quad I_a \in C \quad (1)$$

The shared anchor gives different subsets a common visual reference. To encourage agreement despite changing support views, two subsets C_A, C_B are aligned by a symmetric self-distillation objective [20], [21]:

$$\begin{aligned} \mathbf{p}_A &= h_\phi(S(C_A)), \quad \mathbf{z}_B = \text{sg}[A_{\phi'}(E(C_B))], \\ \mathcal{L}_{\text{agg}} &= D(\mathbf{p}_A, \mathbf{z}_B) + D(\mathbf{p}_B, \mathbf{z}_A) \end{aligned} \quad (2)$$

The terms $\mathbf{p}_B, \mathbf{z}_A$ swap A and B . Here $E(C)$ collects the encoded views, h_ϕ is the online predictor, sg stops gradients, and D is cosine distance between corresponding scene-token arrays. The target parameters follow an exponential moving average (EMA), $\phi' \leftarrow \beta\phi' + (1 - \beta)\phi$ [20]. This stage defines the representation that the deployed model should infer: scene tokens learned from multiple views, rather than features of the current image alone. The trained teacher supplies completion targets S_t^* ; its extra views are used only during learning and offline diagnostics.

B. Anchored Completion with Temporal Memory

Retaining history and incorporating current evidence.

The completion model must infer the multi-view target from a partial observation. A gated recurrent unit (GRU) [22]

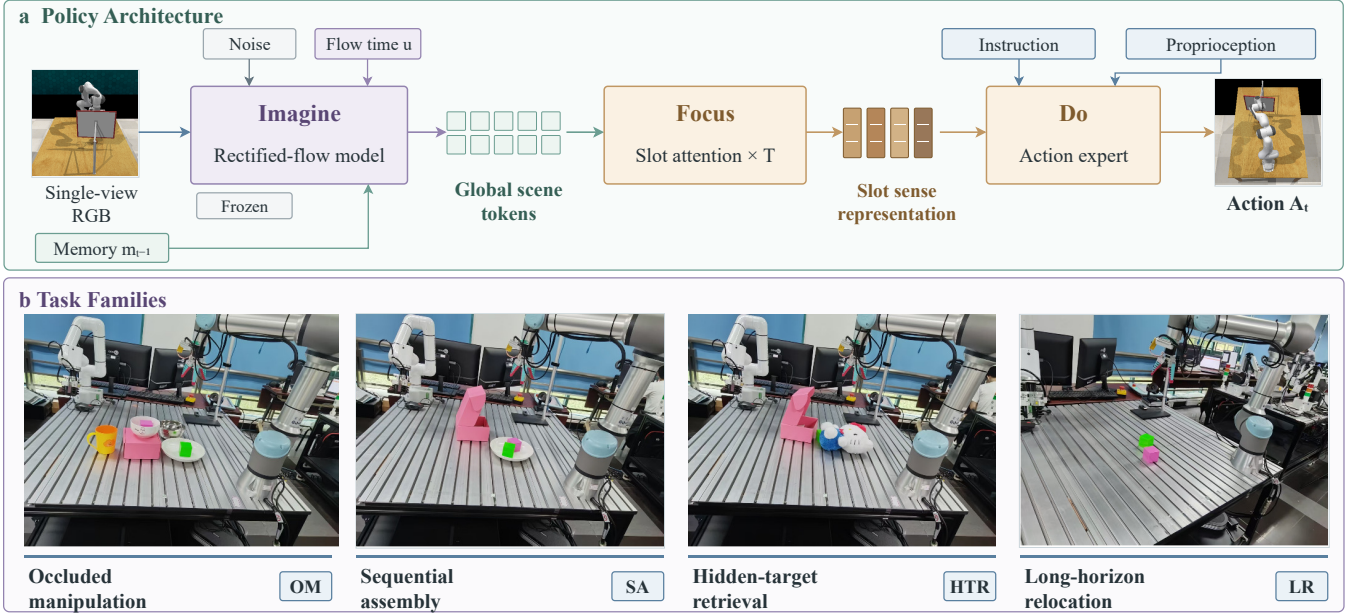


Fig. 2. **Policy and physical tasks.** (a) Imagine completes scene tokens, Focus condenses them into slots, and Do predicts actions. (b) UR5e task families: Occluded Manipulation (OM), Sequential Assembly (SA), Hidden-Target Retrieval (HTR), and Long-Horizon Relocation (LR).

retains observation history, and memory queries attend to current LST:

$$\begin{aligned} \mathbf{m}_t &= \text{GRU}(\mathbf{T}_t, \mathbf{m}_{t-1}), \\ \bar{\mathbf{m}}_t &= \text{Attn}(\mathbf{m}_t, \mathbf{T}_t, \mathbf{T}_t) \end{aligned} \quad (3)$$

Thus history enters completion through features conditioned on the visible scene. This is learned feature alignment, without a known geometric transformation between views. Memory updates once per environment observation; we distinguish environment time t from generative flow progress $u \in [0, 1]$.

Visible-feature anchoring. Historical evidence can inform missing content, but the current view must continue to condition the generated representation. After mapping the two feature arrays to compatible dimensions, FiLM modulation [23] combines them as

$$\begin{aligned} \mathbf{C}_t(u) &= \alpha(u) \text{FiLM}_{\text{vis}}(\mathbf{T}_t) + (1 - \alpha(u)) \text{FiLM}_{\text{mem}}(\bar{\mathbf{m}}_t), \\ \alpha(u) &= \alpha_{\min} + u(\alpha_{\max} - \alpha_{\min}) \end{aligned} \quad (4)$$

With $0 \leq \alpha_{\min} \leq \alpha_{\max} \leq 1$, the weight on current visual evidence is nondecreasing along the flow. Anchor-Guided Cross-Attention (AGCA) supplies this condition to every block of a DiT-style transformer [24]. For evolving hidden tokens $\mathbf{H}$,

$$\mathbf{H} \leftarrow \mathbf{H} + \text{MHA}(\text{LN}(\mathbf{H}), W_K \mathbf{C}_t(u), W_V \mathbf{C}_t(u)) \quad (5)$$

Here MHA and LN denote multi-head attention and layer normalization. Self-attention exchanges information among scene tokens, while cross-attention repeatedly exposes them to $\mathbf{C}_t(u)$. This provides learned anchoring throughout generation, without imposing exact geometric consistency.

Learning to complete scene tokens. The network learns a rectified flow [19] from Gaussian noise $\epsilon \sim \mathcal{N}(0, \mathbf{I})$ to the

teacher representation:

$$\begin{aligned} \mathbf{X}_t(u) &= (1 - u)\epsilon + u\mathbf{S}_t^*, \\ \mathcal{L}_{\text{RF}} &= \mathbb{E} \left[\|v_\theta(\mathbf{X}_t(u), u, \mathbf{C}_t(u)) - (\mathbf{S}_t^* - \epsilon)\|_F^2 \right] \end{aligned} \quad (6)$$

The expectation covers training observations, noise, and sampled flow progress. At inference, integrating the learned vector field from $u = 0$ to $u = 1$, with classifier-free guidance [25], produces $\hat{\mathbf{S}}_t$. The number of integration steps J controls inference cost; within these steps, memory stays fixed while its weighting against visible features changes with u . The generated tokens represent the current scene for control; the recurrence in Eq. (3) accumulates observed LST. Appendix A gives the attention, modulation, and guidance details.

C. Policy Adaptation from Scene Tokens

The Focus module applies slot attention [34] to $\hat{\mathbf{S}}_t$, producing K_s compact slot features $\mathbf{Z}_t \in \mathbb{R}^{K_s \times d}$. Slots compete for scene-token evidence and iteratively aggregate it, providing a compact interface between scene completion and control. Appendix A specifies scene-conditioned initialization and slot updates. This grouping does not require each slot to correspond to one physical object. The Do module predicts an action from slots, proprioception, and language:

$$\hat{\mathbf{A}}_t = \pi_\psi([\text{vec}(\mathbf{Z}_t); \mathbf{P}_t; \mathbf{e}_L]) \quad (7)$$

During Staged policy learning, perception and completion remain frozen, while slot attention and the policy head learn from demonstrations:

$$\mathcal{L}_{\text{policy}} = \mathbb{E}_t \|\hat{\mathbf{A}}_t - \mathbf{A}_t^*\|_1 + \lambda_{\text{tv}} \mathbb{E}_t \|\hat{\mathbf{A}}_t - \hat{\mathbf{A}}_{t-1}\|_2^2 \quad (8)$$

Here $\mathbf{A}_t^*$ is the demonstration action and $\lambda_{\text{tv}} \geq 0$ weights temporal smoothness. The first term imitates actions and the

TABLE I

MAIN RESULTS ON LIBERO, METAWORLD, AND ROBOTWIN 2.0 UNDER THE CORRESPONDING BENCHMARK SETTINGS. LIFD SUITE-LEVEL LIBERO AND METAWORLD ENTRIES REPORT MEAN $\pm$ SE; LIBERO AVG AVERAGES THE FOUR SUITE MEANS. OTHER METHODS REPORT MEANS. SCI ($\downarrow$) IS EVALUATED ON LIBERO-SPATIAL FOR LIFD VARIANTS. $\dagger$: 3D-POLICY ARCHITECTURES ADAPTED TO THE SINGLE-VIEW RGB EVALUATION INTERFACE IN LIBERO AND THE PHYSICAL EVALUATION; ROBOTWIN ENTRIES RETAIN PUBLISHED SETTINGS. $-$: UNREPORTED OR INAPPLICABLE.

Method	LIBERO SR (%) $\uparrow$				MetaWorld		SCI $\downarrow$	RoboTwin 2.0 SR (%) $\uparrow$	
	Spatial	Object	Goal	Long	Avg	Avg. SR (%) $\uparrow$		Hard (Clean $\rightarrow$ Random)	Easy (Clean $\rightarrow$ Clean)
BC-R3M	54.0	51.0	43.0	41.0	47.3	–	–	–	–
VC-1	58.0	56.0	45.0	43.0	50.5	–	–	–	–
PerAct †	61.5	57.8	49.3	47.0	53.9	–	–	–	–
ACT [26]	82.0	78.8	66.1	44.0	67.7	–	–	1.7	29.7
DP3 †	63.2	59.5	51.0	49.1	55.7	–	–	5.0	55.2
Lift3D †	64.0	60.2	51.8	50.0	56.5	–	–	–	–
Diffusion Policy	66.0	62.0	54.0	52.0	58.5	10.5	–	0.6	28.0
OpenVLA-7B	77.5	73.5	64.1	60.3	68.9	–	–	–	–
OpenVLA-OFT [27]	97.6	98.4	97.9	94.5	97.1	–	–	–	–
RDT-1B [28]	60.2	77.8	68.2	29.0	58.8	–	–	13.7	34.5
π_0	79.2	75.0	65.5	61.7	70.4	47.9	–	16.3	46.4
$\pi_{0.5}$ [29]	98.8	98.2	98.0	92.4	96.9	81.6	–	46.0	70.7
GR00T N1.6 [30]	97.7	98.5	97.5	94.4	97.0	–	–	–	–
FabriVLA [31]	–	–	–	–	–	90.0	–	–	–
SUREFlow [32]	94.8	91.0	93.8	90.2	92.5	88.3	–	–	–
LA4VLA-1B [33]	93.4	98.2	98.0	93.4	95.8	87.5	–	–	–
LIFD (Joint)	90.8 $\pm$ 2.4	89.7 $\pm$ 2.7	88.2 $\pm$ 1.9	85.4 $\pm$ 2.2	88.5	77.4 $\pm$ 2.8	0.20 $\pm$ 0.06	16.8	59.6
LIFD (Staged)	93.6 $\pm$ 1.7	92.8 $\pm$ 1.5	91.3 $\pm$ 2.1	88.6 $\pm$ 1.8	91.6	79.8 $\pm$ 1.9	0.14 $\pm$ 0.03	19.4	63.2

second penalizes abrupt changes. Separating this objective from representation learning allows the same scene representation to support downstream policy adaptation.

Joint versus Staged training. GeoVGGT-R remains frozen in both schedules. Staged first learns the multi-view teacher, then recurrent-memory-conditioned RF completion toward that frozen teacher, and finally slots/policy with perception/completion frozen. Joint simultaneously optimizes aggregation, RF/recurrent-memory completion, and slots/policy, combining representation, completion, and control objectives. This tests representation-first against simultaneous optimization.

IV. EXPERIMENTS

Evaluation setting. We evaluate LIBERO-Spatial, Object, Goal, and Long (LIBERO-10) [35] and 50 MetaWorld tasks [36], using 50 demonstrations/task. Multi-view supervision trains the representation; deployed LIFD receives one 224×224 RGB image per step and recurrent history. LIBERO success follows benchmark task criteria. LIFD’s LIBERO/MetaWorld results use three training/evaluation seeds $\times$ 50 rollouts/task/seed (150/task), reporting mean $\pm$ SE; other methods report means only. LIBERO Avg averages the four suite means. Beyond policy demonstrations, GeoVGGT-R uses 100,000 simulated frames from RL Bench, ManiSkill3, and RoboTwin and 10,000 real frames with geometric supervision. Table I summarizes corresponding benchmark settings and is not intended as an input-matched comparison across all methods; Appendix B records settings and sources.

A. Control from Scene Memory

LIFD (Staged) achieves 91.6% LIBERO average success and 79.8% MetaWorld success (Table I). MetaWorld offers complementary validation: LIFD is competitive but 1.8 points below $\pi_{0.5}$ (81.6%) and below the strongest listed references. Staged improves on Joint by 2.8–3.2 points across LIBERO suites, 3.1 on average, and 2.4 on MetaWorld, supporting representation-first learning.

RoboTwin uses 50 clean demonstrations/task and 100 test rollouts/task/condition across 50 tasks [37], without randomized-domain training. Hard (Clean $\rightarrow$ Random) precedes Easy (Clean $\rightarrow$ Clean). Staged achieves 19.4%/63.2% versus Joint’s 16.8%/59.6%, improving by 2.6/3.6 points. Hard tests direct generalization to randomized environments; its 43.8-point decrease from Easy shows remaining sensitivity to this shift.

B. Completion and Observation History

Scene consistency. The Scene Contradiction Index (SCI, lower is better) measures generated-token agreement with a frozen multi-view teacher after Hungarian matching. It is a teacher-relative latent diagnostic, not an independent measure of geometric accuracy; task success remains primary. Table I reports LIBERO-Spatial SCI as mean $\pm$ SE over three training seeds (aggregation: Appendix B); baselines without compatible tokens have no entry. Staged reduces SCI from 0.20 to 0.14 alongside its control gains.

Completion and anchoring. Table II tests LIBERO-Spatial components against the 93.6% full-model result in Table I. FiLM-only (no RF) retains current-view and memory

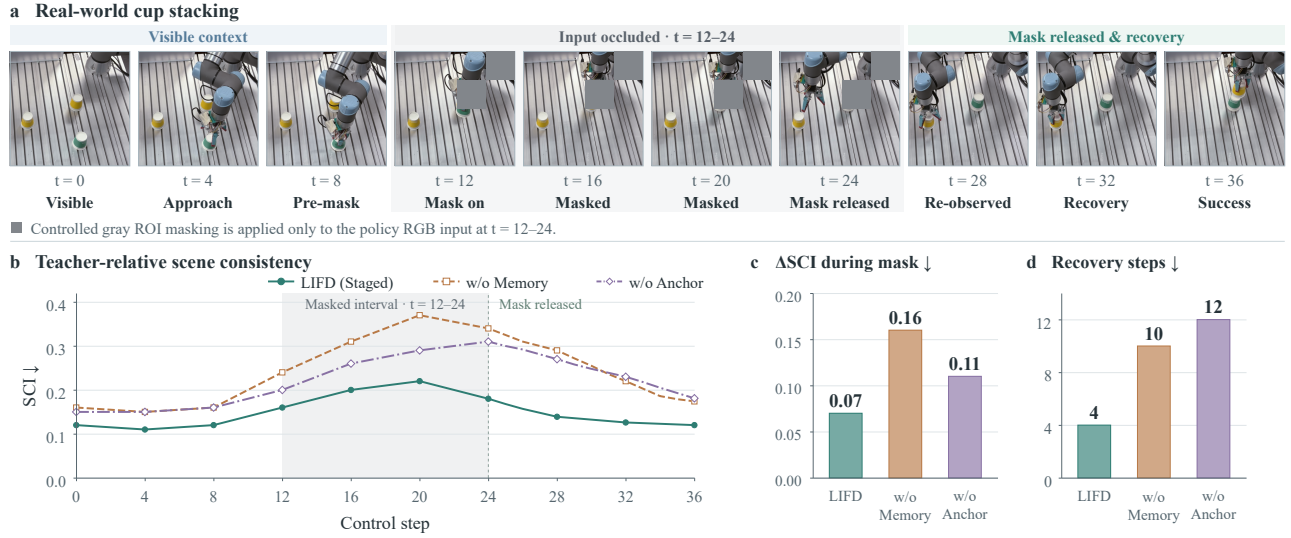


Fig. 3. **Real-world controlled occlusion.** A fixed gray ROI masks only the policy RGB input at steps 12–24 and is released immediately after step 24. (a) Keyframes through recovery and success. (b) Stepwise teacher-relative SCI. (c) Contradiction increase during masking. (d) Recovery steps after release. Summary definitions are in Appendix B.

TABLE II

LIBERO-SPATIAL ABLATIONS. SR: MEAN $\pm$ SE ACROSS THREE SEEDS; SCI: POINT ESTIMATES. DROPS USE THE 93.6% FULL-MODEL CONTROL, AS IN TABLE I.

Variant	SR (%) $\uparrow$	SCI $\downarrow$
Without anchor in AGCA	69.9 $\pm$ 3.9	0.46
FiLM-only (no RF)	30.8 $\pm$ 5.7	0.62
Half-blind teacher	51.1 $\pm$ 5.6	0.24
Without memory in AGCA	74.8 $\pm$ 2.4	0.19
Without slot attention	70.3 $\pm$ 1.3	0.20
Without CFG	65.2 $\pm$ 4.2	0.32
LIFD (Staged)	93.6 $\pm$ 1.7	0.14

FiLM fusion but removes generative completion, passing the deterministic representation directly to scene readout/control. Success falls by 62.8 points and SCI rises from 0.14 to 0.62. Without anchor removes direct current-view conditioning but retains memory-conditioned AGCA (−23.7 points); without memory retains only current-view conditioning (−18.8 points). The Half-blind teacher uses the deployed policy’s same-view evidence instead of richer multi-view evidence (−42.5 points). Removing iterative object-centric slot grouping uses direct, non-slot global policy readout (−23.3 points). Without CFG retains the full model and training-time condition dropout but uses the conditional vector field alone at inference; its 28.4-point decrease supports guidance for reliable completion.

Occlusion and memory interventions. In LIBERO-Spatial, random square occluders cover 0–60% of each image in 10% increments. With identical masks covering at least 40%, LIFD retains an 8–10-point success advantage over OpenVLA-7B. This tests missing-pixel robustness, not hidden-object reconstruction. On LIBERO-Long, zeroing memory at steps 10, 20, or 40 with weights and noise schedules fixed produces the largest reported decrease at step 20.

TABLE III

UR5e SUCCESS (% , $\uparrow$): TEN DEMONSTRATIONS AND 50 TRIALS PER TASK FAMILY. MEAN AVERAGES THE FOUR FAMILIES EQUALLY; $\uparrow$ FOLLOWS TABLE I.

Method	OM	SA	HTR	LR	Mean
BC-R3M	36	12	32	4	21.0
OpenVLA-7B	54	32	64	12	40.5
Diffusion Policy	42	20	46	16	31.0
Lift3D $\uparrow$	48	34	48	8	34.5
LIFD (Staged)	68	62	66	28	56.0

This dependence supports accumulated history’s control utility without identifying its encoded physical quantities.

Fig. 3 applies a fixed gray ROI only to cup-stacking policy RGB at steps 12–24, releasing it immediately after step 24; the first displayed re-observation is step 28. Full LIFD has the smallest masked-period contradiction increase ($\Delta\text{SCI}_{\text{mask}}$: 0.07 versus 0.16 without memory and 0.11 without anchor) and fastest recovery after release (four versus ten and twelve control steps). Memory removal causes the largest degradation during occlusion; anchor removal leaves a longer stale-state tail after observations return. These are teacher-relative consistency/recovery readouts, not metric hidden-geometry reconstruction.

C. Physical Adaptation and Deployment Costs

The physical evaluation uses a UR5e with a laterally mounted Intel RealSense D435i RGB stream across twelve instances in four task families (Fig. 2): occluded-object extraction, ordered insertion/removal, hidden-target retrieval, and relocation under viewpoint changes. Ten human demonstrations per family adapt only the policy head; perception/completion remain frozen after representation learning with real images. Each method receives 50 independent trials/family spanning predefined variations in poses, distrac-

tors, lighting, and camera extrinsics (Appendix B).

LIFD achieves 56.0% macro-average success versus OpenVLA-7B’s 40.5%, a 15.5-point gain (Table III). SA has the largest gain, 30 points; HTR gains only 2 points (one additional success/50 trials). LR remains difficult: LIFD succeeds in 14/50 trials (28%) versus 8/50 (16%) for Diffusion Policy, that task’s strongest baseline.

Inference cost. We use 50 RF integration steps by default. On a single RTX 4090, reducing integration to 35 and 20 steps lowers end-to-end latency from 190 ms to 130 ms and 75 ms, respectively. The corresponding reciprocal throughputs are 5.3, 7.7, and 13.3 policy updates per second, distinct from the robot command-interface rate. Appendix B reports the full timing sweep (Table IV).

V. DISCUSSION AND LIMITATIONS

The results support the control value of combining geometric perception, generative completion, and observation history. LIFD learns this scene representation with additional multi-view supervision and maintains it through iterative inference. Its teacher-relative SCI does not independently establish geometric accuracy, and recovering objects never observed in an episode remains untested. Differing pretraining resources and one physical platform with four task families limit the comparison. Low relocation success identifies a remaining control limitation.

VI. CONCLUSION

We presented LIFD, a framework that connects RGB-based geometric perception to manipulation through persistent, 3D-aware scene memory. Multi-view supervision defines a scene-token representation; anchored generative completion combines current observations with recurrent history, and slot features provide the policy interface. Simulation and UR5e results support the control utility of this representation within the evaluated adaptation settings, while independent verification of hidden-scene geometry remains open.

APPENDIX A IMPLEMENTATION DETAILS

A. Geometric Supervision and Multi-View Aggregation

Fig. 4 shows supervision and reconstructions before/after geometric adaptation. The encoder minimizes

$$\mathcal{L}_{\text{enc}} = \mathcal{L}_{\text{cam}} + \mathcal{L}_{\text{depth}} + \mathcal{L}_{\text{pmap}} + \lambda_{\text{track}} \mathcal{L}_{\text{track}} \quad (9)$$

For view v , camera parameters $\mathbf{g}_v$ and track locations $\hat{\mathbf{y}}_{j,v}$ are supervised by

$$\begin{aligned} \mathcal{L}_{\text{cam}} &= \sum_v \rho_{\delta}(\mathbf{g}_v - \mathbf{g}_v^{\text{gt}}), \\ \mathcal{L}_{\text{track}} &= \sum_{j,v} \|\hat{\mathbf{y}}_{j,v} - \mathbf{y}_{j,v}^{\text{gt}}\|_2, \end{aligned} \quad (10)$$

The Huber loss is $\rho_{\delta}(\mathbf{r}) = \|\mathbf{r}\|_2^2/(2\delta)$ when $\|\mathbf{r}\|_2 \leq \delta$, and $\|\mathbf{r}\|_2 - \delta/2$ otherwise. Depth and point-map heads each use

$$\begin{aligned} \mathcal{L}_{\text{dense}} &= \sum_v \left[\|\sigma_v \odot (\hat{\mathbf{Y}}_v - \mathbf{Y}_v^{\text{gt}})\|_1 \right. \\ &\quad \left. + \|\sigma_v \odot (\nabla \hat{\mathbf{Y}}_v - \nabla \mathbf{Y}_v^{\text{gt}})\|_1 - \lambda_{\text{conf}} \sum_p \log \sigma_{v,p} \right] \end{aligned} \quad (11)$$

Here $\mathbf{Y}$ is depth or point-map data, ∇ the spatial gradient, and p a pixel index. Each head predicts separate softplus weights $\sigma_v > 0$, regularized by $\lambda_{\text{conf}} > 0$.

For support views $\mathcal{I}_{\text{supp}} = \mathcal{I} \setminus \{I_a\}$, curriculum level k admits

$$\begin{aligned} \mathcal{C}^{(k)} &= \{\{I_a\} \cup U : U \subseteq \mathcal{I}_{\text{supp}}, |U| = k-1\}, \\ k &= 2, \dots, n \end{aligned} \quad (12)$$

For encoded subset $\mathbf{F}_C$ and learned queries $\mathbf{U}^{(0)} = \mathbf{Q}_{\text{agg}} \in \mathbb{R}^{M \times d}$, aggregation alternates cross- and self-attention:

$$\begin{aligned} \tilde{\mathbf{U}}^{(\ell)} &= \mathbf{U}^{(\ell-1)} + \text{MHA}(\text{LN}(\mathbf{U}^{(\ell-1)}), K(\mathbf{F}_C), V(\mathbf{F}_C)), \\ \mathbf{U}^{(\ell)} &= \tilde{\mathbf{U}}^{(\ell)} + \text{MHA}(\text{LN}(\tilde{\mathbf{U}}^{(\ell)}), K(\tilde{\mathbf{U}}^{(\ell)}), V(\tilde{\mathbf{U}}^{(\ell)})) \end{aligned} \quad (13)$$

K, V are learned projections, LN is layer normalization, and $\mathbf{S}(C) = \mathbf{U}^{(L_{\text{agg}})}$.

B. Flow Conditioning and Slot Readout

Time MLP f_{time} modulates transformer normalization:

$$\begin{aligned} [\gamma(u), \mathbf{b}(u)] &= f_{\text{time}}(u), \\ \text{AdaLN}_u(\mathbf{H}) &= \gamma(u) \odot \text{LN}(\mathbf{H}) + \mathbf{b}(u) \end{aligned} \quad (14)$$

Scale and shift are broadcast over tokens; each block ends with pointwise feedforward layers. Training drops conditioning with probability p_{drop} . At identical tokens and flow progress, classifier-free guidance combines conditional/unconditional fields:

$$v^{\text{cfg}} = v_{\text{cond}} + w(v_{\text{cond}} - v_{\text{uncond}}), \quad (15)$$

With guidance weight w , integrate $d\mathbf{X}/du = v^{\text{cfg}}$ from noise ($u = 0$) to scene tokens ($u = 1$).

For $\mathbf{x}_i = \hat{\mathbf{s}}_{t,i}$ and $\mathbf{c}_t = M^{-1} \sum_i \mathbf{x}_i$, initialize scene-conditioned Gaussian slots:

$$\begin{aligned} [\boldsymbol{\mu}_0, \log \boldsymbol{\sigma}_0] &= g_{\text{init}}(\mathbf{c}_t), \\ \mathbf{z}_k^{(0)} &\sim \mathcal{N}(\boldsymbol{\mu}_0, \text{diag}(\boldsymbol{\sigma}_0^2)), \quad k = 1, \dots, K_s \end{aligned} \quad (16)$$

Keys $\mathbf{k}_i = W_k \mathbf{x}_i$, values $\mathbf{v}_i = W_v \mathbf{x}_i$, and queries $\mathbf{q}_k^{(\ell)} = W_q \text{LN}(\mathbf{z}_k^{(\ell)})$ allocate evidence:

$$\begin{aligned} e_{ik}^{(\ell)} &= (\mathbf{q}_k^{(\ell)})^{\top} \mathbf{k}_i / \sqrt{d}, \\ a_{ik}^{(\ell)} &= \frac{\exp(e_{ik}^{(\ell)})}{\sum_{j=1}^{K_s} \exp(e_{ij}^{(\ell)})}, \\ \mathbf{r}_k^{(\ell)} &= \frac{\sum_{i=1}^M a_{ik}^{(\ell)} \mathbf{v}_i}{\sum_{i=1}^M a_{ik}^{(\ell)} + \varepsilon_{\text{attn}}} \end{aligned} \quad (17)$$

Softmax normalizes over slots and evidence averaging over tokens, with stabilizer $\varepsilon_{\text{attn}} > 0$. Update slots by

$$\mathbf{z}_k^{(\ell+1)} = G_{\text{slot}}(\mathbf{z}_k^{(\ell)}, \mathbf{r}_k^{(\ell)}) + f_{\text{res}}(\text{LN}(\mathbf{z}_k^{(\ell)})), \quad (18)$$

G_{slot} is a GRU that takes the previous slot state and aggregated evidence, and f_{res} is an MLP; stacking the slots after L_s iterations yields $\mathbf{Z}_t$.

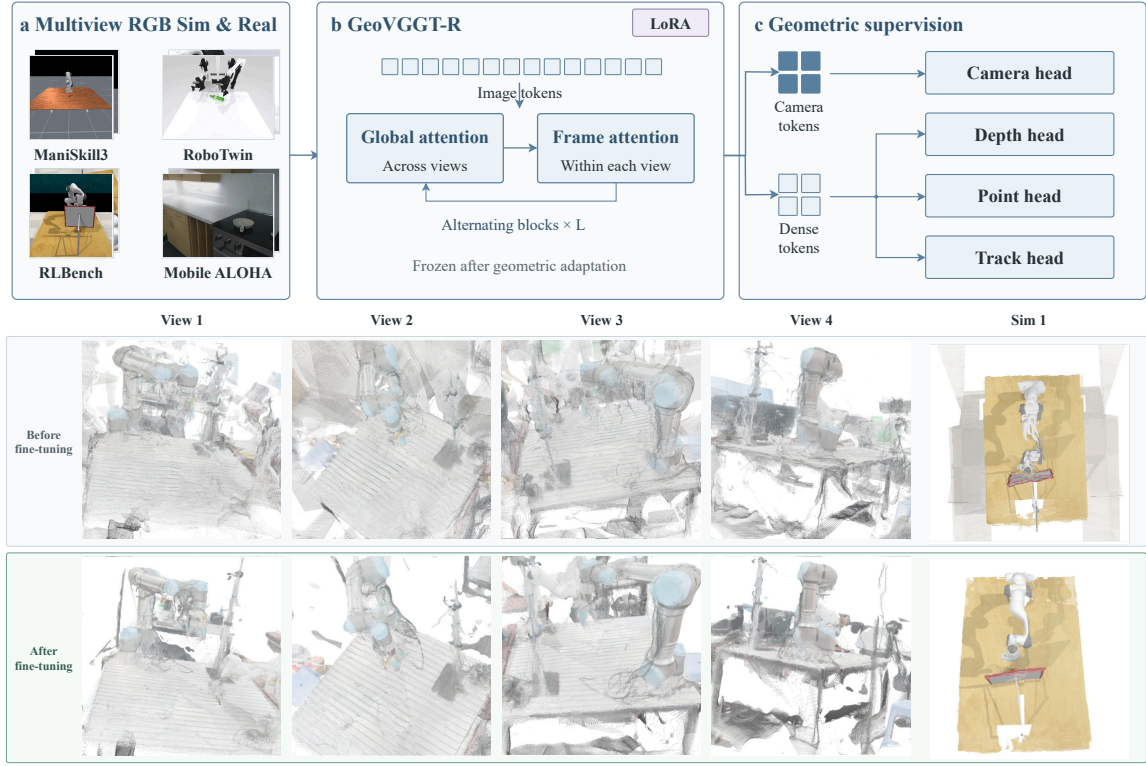


Fig. 4. **GeoVGGT-R**. Top: geometric supervision of camera and dense tokens. Bottom: qualitative reconstructions before and after adaptation from four real views and one simulated scene.

APPENDIX B EVALUATION PROTOCOL AND DIAGNOSTICS

TABLE IV

RTX 4090 INFERENCE COST. UPDATES/S IS RECIPROCAL END-TO-END LATENCY, DISTINCT FROM ROBOT-INTERFACE FREQUENCY.

RF steps	Latency (ms)↓	Updates/s↑
50	190	5.3
35	130	7.7
20	75	13.3

Evaluation counts. Our LIBERO/MetaWorld results use three seeds $\times$ 50 rollouts/task/seed (150/task). LIBERO Avg averages four suites; MetaWorld Avg uses the benchmark’s reported aggregate over difficulty tiers. RoboTwin aggregates 100 rollouts/task/condition over 50 tasks.

Training observations and baselines. Representation training uses seven views: front (anchor), top, back, left, right, front-left, and front-right. Teacher aggregation uses progressively larger anchored subsets and EMA coefficient [0.99, 0.999].

Baselines include pretrained 2D features [10], [11], action diffusion [15], VLAs [38], [39], and adapted 3D policies [1]–[3]. RoboTwin entries retain the sensing and clean-data training settings reported by the official leaderboard [40]. **Result sources.** LIBERO ACT uses STAR’s 50-demonstration/task results [41]; RDT-1B uses LatBot [42]. MetaWorld reference results use Evo-SOTA standard-evaluation averages [43], including LA4VLA-1B MixPT [33]. OpenVLA-OFT uses

external/wrist cameras and filtered demonstrations [27]; LIBERO $\pi_{0.5}$ /GR00T N1.6 scores follow the baseline results reported in ABot-M0 [44]. RoboTwin’s board includes Single-task SFT and Co-train tracks [40]: $\pi_{0.5}$ trains across 50 clean-data tasks, while the other listed public baselines use per-task fine-tuning.

Scene Contradiction Index. A multi-view teacher, trained on training data and frozen for evaluation, provides S_t^* for diagnosis, never as policy input. Hungarian permutation Π_t minimizes generated–teacher cosine distance. With episode length H and warm-up $W = 10$,

$$\text{SCI} = 1 - \frac{1}{H - W} \sum_{t=W+1}^H c_t, \quad (19)$$

$$c_t = \frac{1}{M} \sum_{m=1}^M \cos(\hat{s}_{t, \Pi_t(m)}, s_{t,m}^*)$$

Table I reports LIBERO-Spatial SCI as mean $\pm$ SE over three training seeds. Within each seed, rollout SCI is averaged within each task and then equally across tasks. Table II reports SCI point estimates.

Cup-intervention summaries. Fig. 3 uses stepwise $\text{SCI}_t = 1 - c_t$ before rollout averaging. Let SCI_{pre} be the mean over steps $\{0, 4, 8\}$. $\Delta \text{SCI}_{\text{mask}}$ is the mean at $\{12, 16, 20, 24\}$ minus SCI_{pre} . Recovery is the first post-release control step with $\text{SCI}_t \leq 1.2 \text{SCI}_{\text{pre}}$, with $T_{\text{rec}} = t_{\text{recover}} - 24$. Recovery uses control-step resolution; displayed keyframes are four steps apart.

Physical tasks and hardware. OM extracts a pink block

partly hidden by a ceramic bowl wall, varying occlusion depth, bowl placement, and distractors. SA inserts green then pink blocks into a container and removes them in prescribed order, varying order, container depth, and initial poses. HTR moves a plush toy fully hiding a green target before grasping it, varying occluder type/mass and target placement. LR stacks a green block on a red block under camera-extrinsic shifts of 2–10 cm/5–20°. Success requires target extraction (OM), the correct insertion/removal sequence (SA), occluder movement followed by hidden-target retrieval (HTR), or completion of the relocation/stacking goal (LR).

The UR5e uses a compliant three-finger gripper. The laterally mounted Intel RealSense D435i faces the workspace center and captures 30 Hz RGB; depth is used only for visualization/annotation, never for control. Cartesian position control sends 6-DoF end-effector poses through Real-Time Data Exchange (RTDE) at 125 Hz, independently of policy inference throughput.

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